

MALAHAT, BC, Canada

WMO: 717740

Lat: 48.575N

Lon: 123.530W

Elev: 1200

StdP: 14.07

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	22.4	26.1	9.2	9.2	37.5	12.9	11.1	41.0	16.5	37.1	15.0	36.9	7.2	30	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.4	81.8	62.3	78.4	61.4	75.4	60.4	65.9	76.2	63.6	74.6	61.9	73.0	4.3	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.8	86.3	71.6	59.1	78.2	68.7	57.3	73.1	66.3	31.3	75.9	29.5	74.9	28.2	73.1	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.9	12.8	11.0	DB	19.0	87.8	4.8	3.0	15.5	90.0	12.7	91.7	10.0	93.4	6.5	95.5	(4)
(5)			WB	18.6	66.4	4.1	3.9	15.6	69.2	13.2	71.5	10.9	73.6	8.0	76.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	38.3	38.5	41.7	46.4	53.5	58.5	64.8	65.0	58.9	49.3	41.7	37.4	(6)	
	DBStd	11.40	5.55	5.17	4.92	5.39	6.50	6.27	6.10	5.57	5.78	4.74	5.36	5.04	(7)	
	HDD50	1837	363	323	263	136	34	3	0	0	2	70	252	391	(8)	
	HDD65	5820	828	743	723	557	361	218	79	69	200	487	699	856	(9)	
	CDD50	1678	0	0	5	29	142	256	459	466	270	48	3	0	(10)	
	CDD65	188	0	0	0	0	5	22	73	70	18	0	0	0	(11)	
	CDH74	967	0	0	0	0	16	127	426	338	60	0	0	0	(12)	
	CDH80	182	0	0	0	0	1	24	101	51	5	0	0	0	(13)	
(14) Wind	WSAvg	4.6	4.8	5.0	5.1	4.8	4.5	4.6	4.0	3.9	3.9	4.4	4.8	5.1	(14)	
(15) Precipitation	PrecAvg	50.9	8.6	5.4	4.7	2.7	1.7	1.4	0.9	1.3	1.8	5.5	8.8	8.4	(15)	
	PrecMax	66.7	14.6	14.6	10.1	5.7	3.4	3.5	1.9	4.8	5.0	14.6	18.6	12.6	(16)	
	PrecMin	36.3	3.3	1.2	1.1	0.7	0.4	0.3	0.1	0.1	0.2	1.0	3.4	4.4	(17)	
	PrecStd	8.5	3.0	2.8	1.9	1.3	0.8	0.7	0.5	1.1	1.3	3.0	4.0	2.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.1	51.0	60.4	68.3	76.1	83.2	88.2	84.8	79.8	65.3	55.1	50.8	(19)	
		MCWB	47.2	45.8	47.8	51.4	58.4	60.5	66.5	62.4	59.4	52.4	51.2	48.7	(20)	
	2%	DB	48.6	48.4	54.2	62.6	71.4	77.6	82.6	80.9	75.2	61.2	52.8	47.8	(21)	
		MCWB	45.3	44.0	46.2	49.5	56.2	59.6	63.1	62.4	59.6	52.4	50.1	45.2	(22)	
	5%	DB	46.7	46.6	51.1	58.4	67.8	72.9	78.4	77.8	71.3	58.3	50.6	45.5	(23)	
		MCWB	43.5	43.0	44.7	48.1	54.6	57.7	62.0	61.7	57.7	51.0	48.1	43.2	(24)	
	10%	DB	45.0	44.8	48.5	54.7	64.1	68.9	75.2	74.6	67.9	55.7	48.5	43.5	(25)	
		MCWB	41.9	41.1	43.4	45.8	52.8	56.2	60.7	60.8	56.7	49.5	45.8	41.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.6	48.5	51.3	55.2	60.8	64.0	71.7	73.1	64.6	57.9	53.1	48.4	(27)	
		MCDB	50.5	50.4	57.5	64.9	72.0	77.3	79.9	74.9	74.0	62.2	54.5	49.6	(28)	
	2%	WB	46.4	45.9	48.4	52.2	58.3	61.7	66.7	66.1	62.1	55.2	50.9	45.7	(29)	
		MCDB	48.2	47.8	52.9	60.6	68.1	74.5	77.5	76.0	71.5	58.5	52.4	46.9	(30)	
	5%	WB	44.5	44.1	46.3	49.7	56.4	59.6	63.9	63.6	60.3	52.9	49.0	43.5	(31)	
		MCDB	46.4	46.1	50.4	56.5	65.3	70.7	74.6	74.7	69.0	56.7	50.9	44.8	(32)	
	10%	WB	42.6	42.4	44.6	47.5	54.6	57.7	61.9	61.8	58.5	50.9	46.7	41.8	(33)	
		MCDB	44.6	44.3	48.5	53.5	63.1	67.4	72.9	72.9	66.2	54.7	48.6	43.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	7.5	9.7	12.1	13.8	14.6	16.3	15.4	12.7	9.0	6.9	5.6	(35)	
		MCDBR	7.6	9.1	12.5	16.6	17.9	19.4	19.6	18.6	16.5	12.4	8.3	7.6	(36)	
	5% WB	MCWBR	6.6	6.9	8.4	10.2	10.9	10.5	10.1	9.6	9.1	8.1	7.1	6.7	(37)	
		MCWBR	7.2	7.7	11.4	14.7	16.0	17.2	18.2	17.2	14.6	10.7	7.9	7.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.303	0.300	0.315	0.341	0.357	0.339	0.331	0.335	0.322	0.316	0.308	0.298	(40)	
		taud	2.559	2.589	2.560	2.474	2.445	2.507	2.570	2.553	2.589	2.590	2.552	2.536	(41)	
	Ebn at Noon		248	277	289	289	287	292	293	288	283	266	243	234	(42)	
		Edh at Noon	18	22	27	32	34	32	30	29	26	22	18	17	(43)	
(44) All-Sky Solar Radiation	RadAvg		291	571	890	1354	1734	1893	2067	1743	1240	682	349	234	(44)	
	RadStd		39	77	106	128	137	182	147	115	102	77	55	31	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page